

Programme	BSc. Computer Science				
Course Title	Foundations of Artificial Intelligence				
Type of Course	Vocational Minor				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	A course on Discrete Mathematics is recommended				
Course Summary	This course provides an introduction to the field of Artificial Intelligence covering fundamental concepts, problem solving methods such as search algorithms and heuristics approaches and different knowledge representation techniques. The course addresses the ethical dimensions of AI and their societal impacts.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Able to gain insight into the evolution of key ideas and technologies by exploring the Artificial Intelligence history and its foundational concepts.	U	C	Instructor-created exams / Quiz/Assignment/ Seminar
CO2	Able to acquire knowledge and skills to understand, design, implement intelligent agents to perceive, reason and act within their environments.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO3	Proficiency in various uninformed and informed search strategies along with constraint satisfaction problem solving methods.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO4	Ability to design and implement logical agents and construct ontologies that capture the semantics of a domain, facilitating knowledge representation.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar

CO5	Understand the ethical considerations of AI and their societal impacts and gain insights into the future trajectory of AI by analysing the emerging trends.	U	C	Instructor-created exams/ Quiz/Assignment/ Seminar
CO6	Represent various AI problems using algorithmic approaches and enhance problem-solving skills by visualizing solutions through the utilization of software tools.	U, Ap	C, P	Practical Assignment / Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus

Module	Unit	Contents	Hrs (45+30)	Marks
I	Introduction to AI		11	18
	1	Artificial Intelligence: Definition and Applications	2	
	2	Foundations of Artificial Intelligence	1	
	3	History of Artificial Intelligence, State of the Art	2	
	4	Intelligent Agents: Agents and Environments	1	
	5	The Concept of Rationality, Nature of Environments: Specifying the Task Environment, Properties of Task Environment	3	
	6	Structure of Agents: Agent Programs, Simple Reflex Agent, Model Based Reflex Agent, Goal Based Agent, Utility Based Agent, Learning Agent (Concept Only, No Algorithm required)	2	
II	AI Problem Solving		14	20
	7	Problem Solving Agents (Concept Only), Examples Problems: Toy problems, Real world problems	3	
	8	Solutions for searching: Tree Search and Graph Search and Measuring Problem Solving Performance (Concept Only)	1	
	9	Uninformed Search Strategies: Breadth First Search, Uniform Cost Search, Depth First Search,	4	
	10	Informed search strategies: Greedy Best First search, A* Search, Heuristic Search (Concept Only)	2	

	11	Constrain Satisfaction Problems: Definition, Examples: Map colouring, Job-Shop scheduling	2	
	12	Constraint Propagation: Node Consistency, Arc Consistency, Path Consistency and K-Consistency	2	
III	Knowledge Representation		13	20
	13	Logical agents: Knowledge based agents, The Wumpus world	2	
	14	Logic: Definition, Propositional logic, Syntax and Semantics, Simple Knowledge Base	3	
	15	First Order Logic: Definition, Syntax and Semantic (Models, Symbols and Interpretations, Terms, Atomic Sentences, Complex Sentences, Quantifiers, Equality)	3	
	16	Ontological Engineering: Definition	1	
	17	Categories and Objects: Physical Composition, Measurements, Objects: Things and Stuff, Process, Time Intervals, Fluent and Objects Quantifying Uncertainty (Concept Only)	4	
IV	AI: Philosophical Foundations and Future		7	12
	18	Weak AI: Can machines act intelligently?	1	
	19	Strong AI: Can machines really think?	2	
	20	Ethics and risks of developing Artificial Intelligence	2	
	21	Agent components and architectures	1	
	22	Are we going in the right direction? What if AI succeed?	1	
V	Lab Activities		30	
		1. Identify the various software platforms for AI programming. 2. Identify the various platforms used for No-Code AI (Google Cloud Auto ML, Microsoft Azure AI Buildeer, IBM Watson Studio etc.) 3. Use chatbot platforms like ChatGPT or any other to engage in conversational interactions and understand how natural language processing works. 4. Use online tools or any applications that demonstrate image recognition capabilities. 5. Use online platforms or software that provide		

		interactive AI demos and simulations, such as neural network visualisers or AI-powered character generators. 6. Demonstrate the use of AI-based image editing tools (Actions: Remove objects from images, enhance details, or perform automated retouching.) 7. Utilize AI-powered text summarisation tools like SummarizeBot or Resoomer to generate summaries of lengthy articles or research papers. 8. Use any presentation software like Microsoft PowerPoint or Google Slides to demonstrate AI-driven design suggestions and layout recommendations. 9. Explore AI-based translation tools such as Google Translate or DeepL for translating text between different languages. 10. Assign students to analyse news articles, advertisements, or social media posts using AI technologies. 11. Introduce Students to data visualisation using Tableau Public, an accessible data visualisation tool. 12. Use Google's Teachable Machine platform to create a simple image classification model.		
		Case Study: Provide students with case studies or examples of AI applications in different domains (e.g., healthcare, finance, marketing).		
		Organize demos of AI technologies and applications, such as virtual assistants, autonomous vehicles, facial recognition systems, and recommendation engines.		

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	-	-	-						
CO 2	2	-	2	2	2	2						
CO 3	2	-	2	2	2	2						
CO 4	2	-	2	2	2	2						
CO 5	2	-	2	2	-	-						

CO 6	1	-	1	1	-	-						
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓

References:

1. Stuart Russell, Peter Norvig, “Artificial Intelligence: A Modern Approach”, 3rd Edition, Prentice Hall, 2010.
2. Deepak Khemani, “A First Course in Artificial Intelligence”, McGraw Hill Education, 2017.
3. Elaine Rich, Kevin Knight, & Shivashankar B Nair, “Artificial Intelligence”, McGraw Hill, 3rd Edition, 2009.